

## PUZZLE CORNER

### Rolling coins

Place two coins side to side like this. (If possible, use coins with a milled edge.) Guess what position the head on the left coin will be in if you roll the coin around the top of the other one until it sits on the right. Try and see – you'll be surprised.



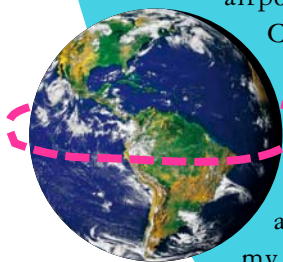
### Bear hunter

A bear hunter leaves his camp and walks 5 miles due south.

Then he turns left and walks 5 miles due east. He spots a bear, but it sees him and charges. He turns left again and runs 5 miles due north, which takes him straight back to camp. What colour was the bear?

### Flying tonight

A woman is sitting crying in an airport lounge on Christmas Eve.



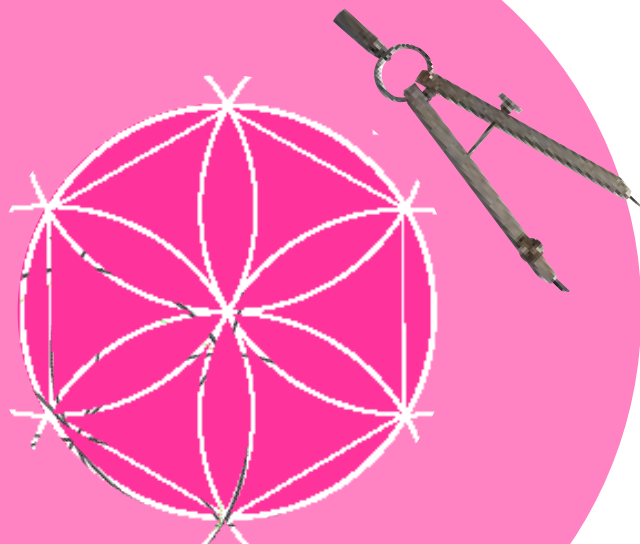
A man walks past and sees her. "What's wrong?" he asks. "I've lost my plane ticket",

she replies, "and now I can't get home for Christmas". "Don't worry", says the man, "I've got my own plane and can give you a lift. I'm going home for Christmas too and can drop you off on the way. It won't add anything to my journey." "But you don't know where I'm going", she replies. "I know", says the man. Where was he going?

# Round and

Try drawing a circle by hand. It's tricky. If you can draw a very good circle, you might have a knack for art. But the way to draw *perfect* circles is by using a pair of compasses. What's more, compasses enable you to make magical designs and drawings.

Follow this pattern to create a hexagon from circles.



## SHAPES FROM CIRCLES

The ancient Greeks were fascinated by circles.

They found that, by using just a pair of compasses and a straight edge (such as a ruler), you can construct lots of other shapes from circles, including hexagons and squares. Here's how to draw a hexagon. Use a pair of compasses to draw a circle. Put the point of the compasses on the circle and draw a curve across it. Move the pin to the crossing point and repeat. Carry on until you've gone right round the circle, then use a ruler to draw between the crossing points.